

2018556059 cshR Berlin Bze

Subject :

Date :

C) SELECT P.Name, P.Price, Count(*) as Number Books, P.PublisherID
FROM PUBLISHERS P, BOOKS B
WHERE P.PublisherID = B.BookID, AND B.BookID =
GROUP BY P.Name, P.PublisherID

D) SELECT B.Title, B.Price
FROM BOOKS B
WHERE B.Price > (SELECT B.Price
FROM BOOKS B1
GROUP BY B1.Price
HAVING COUNT(*) > ALL (SELECT COUNT(*)
FROM BOOKS B2
GROUP BY B2.Price))

E) SELECT BOOKS.Title FROM BOOKS B
WHERE B.Price > (SELECT B.Price FROM BOOKS B1
GROUP BY B1.Price HAVING COUNT(*) > ALL (SELECT
COUNT(*) FROM BOOKS B2 GROUP BY B2.Price))

3) A)

B) R is not BCNF because none B, C, D, E contain a key

1) a) $\Pi_{\text{Title, Price}} \left(\left(\Pi_{\text{AuthorID}} \sigma_{\text{Name} = 'xyz'} (\text{AUTHORS}) \right) \cap \left(\Pi_{\text{PublisherID}} \sigma_{\text{Country} = 'England'} (\text{PUBLISHERS}) \right) \times \text{BOOKS} \right)$

b) $\Pi_{\text{PublisherID, Name, Address}} (\text{PUBLISHERS}) \div \left(\Pi_{\text{PublisherID}} (\text{BOOKS}) \right)$

c) $\left(\Pi_{\text{AuthorID, Name}} (\text{AUTHORS}) \right) - \left(\left(\Pi_{\text{AuthorID}} (\text{AUTHORS}) \right) \times \text{BOOKS} \right)$

2)

a) $\text{SELECT B.Title, B.Price FROM BOOKS B, Authors A, PUBLISHERS P WHERE A.Name = 'xyz' AND P.Country = 'England'}$

b) $\text{SELECT B.Title FROM BOOKS B WHERE NOT EXISTS ((SELECT B2.BookID FROM BOOKS B2 EXCEPT (SELECT A.AuthorID FROM AUTHORS A WHERE A.AuthorID = B2.BookID)))}$